

ROHITH REDDY DEPA

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EXPERIENCE

University of Houston

Oct 2023–Present

Full Stack Developer | Houston, TX

- Developed a modular, scalable UI using **Blazor** and **Razor Components** with **SignalR**, **form validation**, route guards, **state containers**, and **OnAfterRenderAsync** optimization for dynamic rendering.
- Built asynchronous, RESTful **ASP.NET Core Web APIs** with **CQRS** via **MediatR**, integrated **Swagger/OpenAPI**, centralized error handling via **Middleware**, and exposed **gRPC** endpoints for inter-service communication.
- Optimized **SQL Server** operations using **EF Core Code First** with **Redis caching**, improving response time by **50%**, utilizing complex LINQ queries, **eager/lazy loading**, and maintaining transactional integrity with **TransactionScope**.
- Deployed services to **Azure App Service** using **Docker**, integrated **SQL Server** via **Azure SQL**, managed static assets with **Blob Storage**, and implemented **Application Insights** and **Auto-scaling** for monitoring and reliability.

OpenText

Nov 2021–Aug 2023

Software Engineer | Hyderabad, India

- Developed a high-performance full-stack application using **Blazor** and **ASP.NET Core**, reducing **API response time by 35%** with **Redis cache** and improving UI load speed by **40%** via **lazy loading** and pre-rendering.
- Designed modular, responsive UI components using **Blazor Server**, with **SignalR**, **route guards**, and **component-level state management** for real-time data handling and seamless routing.
- Improved **SQL Server** performance by using **indexed views**, **partitioning**, and **query tuning**, increasing report speed by **40%**.
- Built a scalable **ETL pipeline** using **Azure Functions**, **Blob Storage**, and **Azure Service Bus**, reducing processing time by **60%**, while managing parallel batch execution with durable triggers.
- Designed CI/CD workflows using **GitHub Actions** with multistage **YAML**, deployed containerized .NET services via **Docker** and **Helm** on **Azure Kubernetes Service (AKS)**.

TECHNICAL SKILLS

Programming Languages	C#, Java, Python, JavaScript/TypeScript, C/C++, Bash, Shell
Web Development	ASP.NET Core, Angular, React, Redux, Node.js, Blazor, Spring Boot, GraphQL, REST APIs, Razor Pages
Databases	SQL Server, PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, CockroachDB, Firebase, SQLite
Cloud	Azure, AWS, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins
Data	Apache Kafka, AWS Glue, Azure Data Factory, Pandas, NumPy, ETL pipelines, Redis caching
CI/CD & Tools	Git, Azure DevOps, Postman, SonarQube, Jira, Confluence, Visual Studio, VS Code, PyCharm, Eclipse

EDUCATION

University of Houston

May 2025

M.S. in Computer Science | GPA: 3.67/4.0

Relevant Coursework: Cloud Computing, Machine Learning, Artificial Intelligence, Database Management, Computer Architecture, Digital Image Processing.

PROJECTS

School Management Software

- Architected a school management platform using **Spring Boot** and **React** on **AWS**, implementing **JWT/OAuth2** for secure authentication, **REST APIs** for seamless communication, and **MySQL** for reliable data storage.
- Implemented **Redux** for state management, **role-based access control (RBAC)** for security, and **Docker** for scalable deployment. Integrated **Razorpay** for online payments, processing **1,000+ monthly transactions**.

ISP Messaging System

- Engineered a scalable messaging system using **Apache Kafka** for real-time microservice communication.
- Integrated **OpenAI API** for query resolution, reducing customer support workload by **40%** through automated responses.
- Utilized **PostgreSQL** for persistent storage and **Docker** to containerize services, enabling seamless deployment.

Traffic Asset Defect Detection for TxDOT

- Designed an **AI-powered computer vision system** using **YOLOv5** with **transfer learning** and **OpenCV**-based preprocessing to detect traffic asset defects, improving detection accuracy by **25%**.
- Automated defect reporting pipeline using **Python scripting** and scheduled batch processing, reducing manual inspection and documentation time by **40%**.